

```
if( gender == 'F' & ++age >= 65 )
```

A Logical Boolean AND [&] works exactly like a Logical AND [&&] with one exception. It will always check both halves of the equation, even if the first is false.

2.3.5 Logical inclusive Boolean OR

```
if( gender == 'F' | age >= 65 )
```

Again, this works just like the Logical OR, but you are guaranteed that both sides of the expression will always be executed. If either half is true, the entire 'if' is true.

2.3.6 Logical exclusive Boolean OR

```
if( gender == 'F' ^ age >= 65 )
```

This Logical exclusive Boolean OR is true only if one side is true and the other false. If both sides are true, the entire expression is false.

2.4 Control structures

2.4.1 The "if" Statement

This control statement is used for decision making in any Java application. The if in Java exactly mirrors C/C++, and it has three variants:

```
if( expression )
    statement;

if( expression )
    statement;
else
    statement;

if( expression )
    statement;
else if( expression )
    statement;
else
    statement;
```

The "expression" must be something that uses the comparison operators and resolves to either true or false. The statement is executed if the expression is true. Only one statement can be made conditional without brackets. If you wish to conditionally execute more than one statement, you use brackets to create a block. If the "expression" is not true, the else branch executes.